

Canonical Foundations of the Methane Metauniverse (MMU)

Geometric–Elastic Principles, Scope, and Methodological Status
Canon Version 2.0

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Abstract

This document defines the canonical foundations of the Methane Metauniverse (MMU). It replaces and extends the previous canonical statement (Version 1.0, DOI: 10.17605/OSF.IO/QDU67) and constitutes the authoritative reference for all MMU–based analysis, simulation, publication, and AI–assisted reasoning.

The Canon specifies the physical assumptions, non–assumptions, definitions, projection principles, scaling laws, and interpretational rules of the MMU framework. It further clarifies the methodological scope of recent MMU publications, including epistemic audits and AI–augmented development, which do not modify the physical foundations defined herein.

The MMU is formulated as a geometric–elastic substrate from which established physical theories emerge as effective, scale–dependent projection descriptions. This Canon is normative and fixed; any future extensions or revisions must be explicitly declared.

1 Purpose and Status

The Methane Metauniverse (MMU) is a geometric–elastic framework intended to identify the structural origin of physical laws. It is not designed to replace established theories such as quantum mechanics, quantum field theory, or general relativity, but to explain why their mathematical structures and empirical success arise.

This Canon serves as a binding reference contract. It defines what is assumed, what is derived, what is explicitly not assumed, and how MMU statements are to be interpreted. All MMU publications, simulations, and AI–assisted analyses are required to remain consistent with the definitions and constraints specified herein.

Canon Version 2.0 consolidates the MMU project following the publication of foundational, phenomenological, and methodological studies. In particular, it clarifies the status of meta–level and AI–related works, which are canon–constrained but do not introduce new physical postulates.

The Canon is intentionally conservative. It freezes the physical foundations of the MMU at their current state and provides a stable reference point for critical evaluation, reproduction, and potential falsification.

2 Relation to Established Physics

The MMU does not contradict experimentally validated theories such as quantum mechanics, quantum electrodynamics, or general relativity within their tested domains of applicability. These theories are interpreted as highly successful effective descriptions that emerge from a deeper geometric structure.

Within their respective regimes, these theories remain valid and operational. The MMU provides an underlying geometric explanation for their operators, constants, and structural forms.

3 Scope and Limitations

The MMU is currently not optimized for highest numerical precision. Its primary strength lies in explanatory power: revealing the geometric origin of operators, constants, scaling laws, and coupling patterns.

Precision matching with quantum electrodynamics or high-order perturbative methods is not the immediate objective. Precision improvements are regarded as a secondary and future goal.

4 Fundamental Assumptions

Spacetime is modeled as a discrete, fixed lattice of identical dual-tetrahedral UR-cells. The lattice nodes do not translate or drift. All observable phenomena arise from internal elastic deformations of these cells.

No additional spatial dimensions, background fields, or fundamental particles are postulated beyond this geometric structure.

5 Internal Degrees of Freedom

Each UR-cell possesses four internal coordinates:

- w_1 : projection axis associated with observable time, energy, and frequency,
- w_2 : electric or radial deformation mode,
- w_3 : torsional deformation mode associated with spin and magnetism,
- w_4 : volumetric deformation mode associated with inertia and gravitation.

The internal axes (w_2, w_3, w_4) are geometrically non-orthogonal, generating intrinsic elastic cross-couplings fixed by the dual-tetrahedral geometry.

6 Elastic Dynamics

For small internal deformations, the dynamics of a UR-cell are governed by a linear elastic equation derived from a quadratic Lagrangian:

$$M\ddot{\mathbf{w}} + K(a)\mathbf{w} = 0,$$

where $\mathbf{w} = (w_2, w_3, w_4)^T$, M is an effective inertial matrix, and $K(a)$ is the elastic stiffness matrix

$$K = \begin{pmatrix} k_2 & k_{23} & k_{24} \\ k_{23} & k_3 & k_{34} \\ k_{24} & k_{34} & k_4 \end{pmatrix}.$$

7 Universal Scaling Law

All elastic stiffnesses obey a universal geometric scaling law with the internal tetrahedral edge length a :

$$k_i(a) \propto a^{-3}.$$

This scaling law constitutes the central quantization mechanism of the MMU and underlies hydrogenic spectra, fine structure, torsional splitting, and gravimetric response.

8 Physical Identification of Elastic Modes

The diagonal stiffnesses admit closed physical identifications:

$$k_2(a) = \frac{2e^2}{4\pi\epsilon_0 a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad k_4(a) = \frac{2Gm^2}{a^3}.$$

These correspond respectively to electric (Coulomb), torsional (spin), and volumetric (gravitational/inertial) elastic responses.

9 Projection Principle

Observable physics arises exclusively through projection of internal motion onto the axis w_1 . Time, energy, frequency, and relativistic effects are manifestations of this projection.

In the continuum limit:

- radial collective response reproduces the Schrödinger equation,
- torsional duality yields spinorial structure and the Dirac equation,
- volumetric deformation produces the weak-field Einstein equation.

10 Emergence of Lorentz Symmetry

In the MMU framework, Lorentz symmetry is not postulated as a fundamental axiom, but arises as an *unavoidable effective property* of the underlying discrete geometric-elastic spacetime structure.

The dynamics of each UR-cell are governed by a quadratic elastic Lagrangian on a dual-tetrahedral lattice. In the continuum limit, the isotropic second-moment structure of the tetrahedral geometry enforces a quadratic dispersion relation

$$\omega^2 = \omega_0^2 + c^2 k^2 + \mathcal{O}(k^4),$$

which is locally Lorentz invariant.

Observable time dilation and length contraction emerge as *projection effects* of internal motion along the transverse coordinates (w_2, w_3, w_4) onto the observable axis w_1 . Transverse internal oscillations reduce the effective projection onto w_1 , yielding the standard relativistic kinematic factors.

Lorentz symmetry in the MMU is therefore *effective and scale-dependent*. Deviations are expected only when the wavelength approaches the internal lattice scale a , with leading corrections scaling as $\mathcal{O}((a/\lambda)^2)$.

Relativistic kinematics thus appears as the necessary low-energy approximation of a discrete elastic spacetime geometry, rather than as an independent fundamental principle.

11 Prestress, Time, and Gravitation (Canonical Statement)

Canonical statement. In the Methane Metauniverse (MMU), spacetime is described as an elastic lattice with internal deformation modes. A homogeneous *volumetric prestress* along the w_4 axis defines the *global time scale* of the universe and appears observationally as the cosmological constant Λ . This prestress is a property of the spacetime ground state and is not interpreted as vacuum energy.

Time. Time is not a fundamental coordinate in the MMU. It emerges from internal dynamical motion projected onto the observable axis w_1 . The existence of time requires motion, not prestress. The *rate of time* is set by the global w_4 prestress, while *local gravitational time dilation* corresponds to spatial variations $\delta w_4(x)$ induced by mass.

Time direction. The arrow of time is not determined by volumetric prestress. A homogeneous *torsional prestress* along the w_3 axis introduces a global chirality of spacetime. In the presence of dissipation, this torsional bias selects a preferred temporal orientation and thereby defines the experienced time direction.

Gravitation. Gravitation is the elastic response of the spacetime lattice to local volumetric deformation. In the continuum limit, this response reproduces Einstein gravity. No additional fields or free constants are introduced.

Scope. This construction constitutes a canonical interpretational refinement of the MMU framework. It preserves all experimentally validated limits of General Relativity while providing a unified mechanical interpretation of time scale, time direction, gravitation, and the cosmological constant.

12 Local Scale–Filtered Interfaces and Atomic Short–Range Structure

Canonical statement. Atomic short–range effects in the Methane Metauniverse (MMU) arise from a *local, finite spacetime deformation* formed by the overlap of interacting UR–cells. This deformation defines a physically real interface region, termed the *S–node*, and cannot be represented as a point–like or contact interaction.

S–node definition. The S–node is the localized overlap region of two or more UR–cells in which their internal elastic degrees of freedom couple. Within this region, the relative volumetric deformation mode w_4 dominates, while the center–of–mass motion remains stationary. The S–node is not a particle, not a field excitation, and not an additional force carrier.

Scale–filtered interface. Experimental data from muonic hydrogen exclude point–like short–range couplings. Consistency therefore requires that S–nodes act as *scale–filtered interfaces* with a finite characteristic range ξ . No new fundamental length scale is introduced.

Geometric scaling law. Dimensional and geometric consistency uniquely constrain the interface range to scale inversely with the reduced mass μ of the interacting system,

$$\xi(\mu) \propto \mu^{-1}.$$

This scaling ensures finite and stable short–range contributions across hydrogenic systems and provides a natural suppression of ultra–short distance effects.

Physical meaning of reduced mass. Within the MMU framework, the reduced mass μ is identified as the effective inertia of the collective volumetric deformation of the S-node. Reduced-mass scaling is therefore not a kinematic artifact but a direct consequence of spacetime elasticity.

Unified ordering of atomic corrections. Hyperfine structure, Lamb shifts, and effective screening effects correspond to successive orders of elastic response within the same S-node interface. These effects must not be modeled independently, as they originate from a single geometric mechanism.

Relation to established physics. This construction does not replace quantum electrodynamics. It provides an ontological explanation for why short-range corrections remain finite and well-behaved in precision spectroscopy. General-relativistic curvature is not invoked at atomic scales; the deformation described here is local, elastic, and internal to the MMU lattice.

Scope. The present canonical statement applies to hydrogenic two-body systems. Multi-nucleon nuclei require multiple interacting S-nodes and are outside the scope of this section.

13 Methodological Stance

The MMU prioritizes geometric consistency, minimal assumptions, and parameter reduction. Predictions must follow from internal structure and scaling laws, not from ad-hoc fitting or phenomenological insertion.

14 Assumptions and Non-Assumptions

Explicit assumptions include discrete dual-tetrahedral UR-cells, projection onto w_1 , universal a^{-3} stiffness scaling, and intrinsic geometric coupling of internal axes.

Explicit non-assumptions include continuous spacetime at the fundamental level, point-like particles, independent quantum postulates, additional dimensions, or ontological primacy of wavefunctions or fields.

15 Interpretational Clarifications

Within the MMU:

- particles are stable standing deformation modes,
- fields are collective lattice responses,
- the wavefunction encodes two real elastic quadratures,
- forces are projection effects of elastic gradients,
- measurement corresponds to physical decoherence.

16 Scientific Intent

The MMU aims at conceptual unification by identifying a single geometric-elastic substrate underlying quantum, electromagnetic, and gravitational phenomena. It is asserted to be structurally explanatory rather than immediately more precise.

17 Human–AI Collaboration

AI tools support algebraic manipulation, consistency checking, and simulation. Conceptual responsibility, physical interpretation, and validation remain human–led.

18 Canonical Status

This document defines the canonical MMU position. All MMU-based publications, simulations, and AI systems are expected to respect the assumptions, principles, scope, and limitations stated herein.

19 MMU Article Corpus and Scope of Established Results

The Methane Metauniverse (MMU) framework is supported by a coherent series of peer-accessible preprints. Each article establishes a specific structural element, derivation, or testable consequence of the MMU geometry. The following list defines the canonical MMU corpus and its validated conceptual scope.

- **MMU1** — Introduces the dual–tetrahedral geometric concept of space, establishing the discrete elastic lattice as a unifying substrate from atomic to cosmic scales.
- **MMU2** — Demonstrates how atomic masses and ionization energies emerge from elastic space response, linking spectra to geometric stiffness.
- **MMU3** — Provides a geometric derivation of the 21 cm hyperfine line as a torsional spin–flip mode of the MMU lattice.
- **MMU4** — Determines the elastic cross–coupling constants k_{23}, k_{24}, k_{34} from Larmor, Zeeman, and Lamb resonances.
- **MMU5** — Fixes the geometry of the UR–tetrahedron and connects elastic constants to lepton scaling relations.
- **MMU6** — Introduces the dissipative MMU field equation, explaining thermal damping and spatial healing as geometric relaxation processes.
- **MMU7** — Documents the methodological framework of AI–assisted theoretical physics within the MMU project.
- **MMU8** — Establishes geometric quantization of the internal edge length a as the origin of discrete spectra.
- **MMU9** — Develops MMU 2.0, unifying internal spin, torsion–volume coupling, and magnetic g –factors.
- **MMU10** — Derives the Planck scale from tetrahedral geometry without introducing independent fundamental units.
- **MMU11** — Identifies an electro–gravity fixed point beyond the Planck scale, linking electric and gravitational stiffness.
- **MMU12** — Translates length–dependent stiffness laws into mass–dependent formulations with leptonic predictions.
- **MMU13** — Formulates the projection principle for quantum orbitals, deriving angular momentum, spin, and Pauli exclusion geometrically.

- **MMU14** — Derives the Dirac equation as an unavoidable external projection of the MMU UR-cell, including antiparticles.
- **MMU15** — Provides a geometric origin of the Rydberg constant from space quantization.
- **MMU16** — Derives ε_0 , μ_0 , Z_0 , and c directly from tetrahedral geometry.
- **MMU17** — Explains lepton masses, Koide geometry, and $(g - 2)$ patterns as torsional eigenmodes.
- **MMU18** — Models the proton as a hexagonal chair rotor and derives its internal eigenmodes without free parameters.
- **MMU19** — Predicts gravitationally induced spectral shifts as a testable consequence of space compression.
- **MMU20** — Demonstrates the emergence of Einstein gravity from microscopic MMU lattice kinematics.
- **MMU21** — Consolidates MMU 3.0 as a minimal geometric-elastic spacetime model unifying quantum, relativistic, and electromagnetic structure.
- **MMU22** — Establishes the geometric necessity of antimatter as a stability mechanism of the dual-tetrahedral lattice.
- **MMU23** — Predicts correlated Balmer, Zeeman, and Lamb isotope shifts from internal length compression.
- **MMU24** — Explains the neutron-proton mass gap as an electro-gravity phase transition.
- **MMU25** — Derives the proton g -factor from chair-quark torsional geometry.
- **MMU26** — Explains lepton masses from torsional geometry and verifies them via hyperfine and isotope data.
- **MMU27** — Interprets black hole horizons and Hawking temperature as projection effects rather than singularities.
- **MMU28** — Shows that geodesics and gravitational redshift arise as projection necessities of the MMU lattice.
- **MMU29** — Demonstrates that the Dirac operator is unavoidable given dual-tetrahedral geometry.
- **MMU30** — Interprets wavefunction linearity and quantum measurement as lattice dynamics of the MMU spacetime structure.
- **MMU31** — Derives quantum dynamics, Lorentz symmetry, and weak decay rates as emergent consequences of geometric-elastic lattice motion and torsional projection in the MMU spacetime structure.
- **MMU32** — Interprets the cosmological constant, gravitational time dilation, and the arrow of time as manifestations of volumetric and torsional prestress states of the elastic MMU spacetime lattice.
- **MMU33** — Interprets atomic binding, hyperfine structure, and the Lamb shift as consequences of a local, scale-filtered spacetime deformation (S-node) in the elastic MMU lattice, providing a geometric origin for reduced-mass scaling and excluding point-like short-range couplings.

- **MMU34** — Systematically reformulates standard foundational objections to discrete spacetime models as explicit, potentially falsifying criteria within the MMU framework, and demonstrates how issues of Lorentz covariance, conservation laws, quantization, and empirical testability can be resolved through well-defined projection principles, lattice dynamics, and continuum limits.
- **MMU35** -Investigates whether AI-human collaboration can support the disciplined development of a genuinely new physical framework, using the MMU as a stress test to evaluate controlled theory formation, documentation, and independent multi-AI epistemic audits under strict constraints designed to suppress hallucination, conceptual drift, and false authority.

This corpus defines the established conceptual and mathematical foundation of the Methane Metauniverse framework. Subsequent MMU developments are expected to be consistent with these results.

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